TRUNKBELAY · TRACEABILITY MATRIX

TrunkBelay traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

TrunkBelay traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 17 user requirements, 18 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Collar holds on a palm trunk	TKB-CAL-001, TKB-BLD-001, TKB-STM-001	Analysis (TKB-CAL-001 B, C), TRL 3; proof test on cut trunk sections at 3.5 kN and a test to failure of a sample frame, TRL 4 (TKB-DEC-001 O7, decided by Amish 2026-10-25)	Met on paper; the friction on wet bark, settlement (10 mm, 14 to 18 mm with pad shear) and the strength margin (about 5.1:1, 4.5:1 with an unequal chain split; the stronger frame is the fallback if the test to failure shows less than 5:1, O7) are to verify at TRL 4 (locking factor 1.54)
UR-002	PR-002	Holds when the load direction changes	TKB-CAL-001, TKB-BLD-001, TKB-STM-001, TKB-OPT-001	Drop and transition tests on trunk sections, the upward case as a misuse test only, TRL 4	At risk for an upward pull: the single top chain cannot hold it. Decided by Amish (16A): the drill rule stays (collar at least 300 mm above the victim's attachment); the double-acting collar is the recorded fallback
UR-003	PR-003	Fits common palms	TKB-CAL-001, TKB-BLD-001, TKB-STM-001	Analysis of the model on 200, 300 and 450 mm trunks (TKB-CAL-001 E2) and of the lock on a leaning trunk (TKB-STM-001 S31), TRL 3; fit trials on surveyed palms and the drill on a leaning palm, TRL 4	Met by design for diameter: the V and the 110-link chain fit the range, 15 links to spare on 450 mm; lean: locking factor 1.21 at 20 deg on the adverse side on the assumed friction, to confirm by test
UR-004	PR-004	Quick to rig	TKB-CAL-001, TKB-BLD-001, TKB-STM-001	Analysis (TKB-CAL-001 E3), TRL 3; timed drills at 3 m height, TRL 4	At risk: about 4.5 minutes estimated, a thin margin
UR-005	PR-005	Controlled lowering	TKB-CAL-001, TKB-BLD-001, TKB-STM-001	Analysis (TKB-CAL-001 D1), TRL 3; lowering tests with a test mass, TRL 4	Met on paper: about 80 N at the hand with the brake carabiner
UR-006	PR-006	Rope length	TKB-CAL-001, TKB-BLD-001	Inspection and analysis (TKB-CAL-001 E1), TRL 3; measured, TRL 4	Met: 30 m of rope, 28 m needed
UR-007	PR-007	Portable	TKB-CAL-001,		Met on paper: 4.67 kg carried; the 3.7 kg bag is hauled up on

User requirement	Product requirement	Title	Evidence	Verification	Status
			TKB-BLD-001	Analysis (TKB-CAL-001 F2), TRL 3; weighing and climbing trial, TRL 4	the tag line (18C); whole kit 8.4 kg
UR-008	PR-008	No damage to the palm	TKB-CAL-001, TKB-BLD-001, TKB-STM-001, TKB-OPT-001	Analysis (TKB-CAL-001 G1; TKB-STM-001 S19, S20), TRL 3; inspection after trials, TRL 4	Met on paper, at risk on green wood: pads 1.9 MPa mean (about 3.8 MPa at the lower edge) against 9.3 MPa air dry, 4.6 to 5.6 MPa green; bare links mark about 0.3 to 1.3 mm (TKB-STM-001 S19 to S21)
UR-009	PR-009	Cost	TKB-CAL-001, TKB-BLD-001	Costed bill of materials (TKB-CAL-001 H2), TRL 3	Met on paper: USD 828 per kit (USD 22 under)
UR-010	PR-010	Teachable	TKB-CAL-001, TKB-BLD-001	Training trial with the partner programme, TRL 4	To verify at TRL 4: cannot be shown on paper
UR-011	PR-011	Holds if the rescuer lets go	TKB-CAL-001	Inspection of the bought descender's rating and certificate (BOM line 17; EN 341 class A or EN 12841 type C), TRL 3; dummy drill, TRL 4	Met by design
UR-012	PR-012	Head up once freed	TKB-CAL-001	Inspection (BOM lines 16, 19, 20; TKB-STM-001 S22), TRL 3; dummy drill, TRL 4	Met by design; fitting time on a head-down dummy to verify at TRL 4
UR-013	PR-013	Survives the climate	TKB-CAL-001	Analysis (TKB-STM-001 S25, S26), TRL 3; inspection after the trial season, TRL 4	To verify at TRL 4 (corrosion and ageing are plausible, not shown)
UR-014	PR-014	Wear parts replaceable	TKB-CAL-001	Inspection of the build plan (TKB-BLD-001), TRL 3	Met by design
UR-015	PR-015	Recognised rescue gear	TKB-CAL-001	Inspection of the BOM and the marking on the spine (TKB-BLD-001 sheet TKB-DWG-105), TRL 3	Met by design
UR-016	PR-016	Made locally	TKB-CAL-001	Inspection of the build plan (TKB-BLD-001), TRL 3	Met by design (an AC TIG welder is needed)

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-017	PR-017	Patent design-around	TKB-CAL-001	Inspection of the drawings and the precis (TKB-PRC-001), TRL 3	Met by design
UR-004	PR-018	Fixed end captive	TKB-CAL-001, TKB-BLD-001	Inspection of the model (cad/fidelity.json, no interference) and the build plan step 5, TRL 3	Met by design: grade 80 connecting link through the 10 mm tether hole (TKB-DEC-001 D-S1)

Gaps

None.